

Alec Lavoie

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I am a person that cares—and I want to work with other people that care. People are willing to pay, and pay extra, for software that is awesome. If you are someone who cares, I'd love to work together.

Experience

Istari Digital

Remote

SENIOR DESIGN ENGINEER • AEROSPACE INTEGRATION & SIMULATION PLATFORM

Apr 2025 – Jun 2026

- Modernized the enterprise product experience by migrating roughly 75% of the application to a new design system—branding, application chrome, layout primitives, and dark mode—via a Strangler-pattern deprecation of the legacy system
- Solved cross-subcontractor collaboration in large defense programs by enabling teams to decompose complex projects into independently versioned primitives shared across permission boundaries, via composable subsystems, an IDE-style workspace, and system-scoped documents, shipping capabilities in time for the upcoming AIAA AVIATION Forum demonstration
- Delivered platform performance improvements in time for the Connecticut Aerospace & Defense Conference (CADC) and beyond—halving client request volume via pagination and lazy hydration for the demonstration, then roughly 3x'ing warm-cache responses on a major list endpoint (700ms to 220ms) via permission caching
- Accelerated development velocity by reducing CI runtime from ~15 to 4–6 minutes through parallelized tests and merged coverage
- Stabilized client performance under connection and read pressure constraints with an architecture-specific custom sync implementation that incrementally patches views from the Registry service, avoiding expensive list requests and permissioning database load
- Secured enterprise adoption with a major defense contractor via a Confluence integration for viewing secure platform resources in customer documentation hubs

Dispense

Remote

SENIOR SOFTWARE ENGINEER • E-COMMERCE STOREFRONT BUILDER

July 2022 – Feb 2023

- Empowered customers to dynamically manage their product catalog pricing by engineering a customizable discount and promotion workflow
- Securely managed and distributed keys and secrets with a fingerprint-based CLI tool
- Addressed degrading NoSQL data health by conducting an in-depth data integrity audit using schema introspection tools, and architected a forward-looking reconciliation strategy

Indigo Agriculture

Remote

FRONT-END SOFTWARE ENGINEER • TRANSPORT MANAGEMENT SYSTEM

July 2020 – Jan 2022

- Rolled out fully featured multi-sided logistics marketplace for agriculture product transportation
- Documented and resolved page-by-page application lighthouse warnings, critical workflow frame jank, gql cache misses, and slashed dependency graph bloat by 60%

Appcues

Boston, MA

SOFTWARE ENGINEER • FLOW BUILDER CHROME EXTENSION

July 2019 – April 2020

- Significantly improved application boot time and reliability by orchestrating a seamless migration from Firebase to PostgreSQL using a dual-read-write shim API

General Indemnity Group

Boston, MA

SOFTWARE ENGINEER • SELF-SERVICE SURETY

May 2018 – July 2019

- Unveiled a mobile-optimized, high-performance e-commerce storefront to feature a comprehensive bond product catalog
- Slashed operational costs by 40% and bolstered content scalability through the design and integration of a headless CMS and a GUI admin portal for the e-commerce site

AspenTech

Bedford, MA

SOFTWARE DEVELOPER • UNIVERSAL PROCESS INDUSTRY MANAGEMENT SYSTEM

Nov 2016 – Apr 2018

- Allowed customers to explore relationships between oil properties by designing and releasing an n-dimensional interactive parallel coordinate chart supporting tens of thousands of data points backed by a Canvas based render queue

Skills

Typescript, React.js, Node.js, Express, Elixir, Machine Learning, NLP, Next.js, Remix.js, Electron, PostgreSQL, MongoDB, GraphQL, REST, Cypress, testing-library, Jest, Mocha, HTML, CSS, Javascript, Figma, Accessibility, tailwind, XState, Redux, Canvas, Webpack, CI/CD, Chrome Extension

Education

Jan 2024 **Machine Learning: Regression and Classification**, DeepLearning.AI, Stanford University

Remote

Jun 2016 **BSc. Computer Science**, University of Massachusetts Dartmouth

Dartmouth, MA